VIT-AP UNIVERSITY

Amaravati, Andhra Pradesh, India

School of Computer Science and Engineering

Implementation and Coding Report

FITPULSE

Detailed Development and Production Readiness Documentation

Course	Software Engineering
Assessment Component	5 - Implementation and Coding
Maximum Marks	10
Allocated Marks	10
Submission Date	14 April 2026
Implementation Baseline Commit	8f7f44f
Technology Stack	React 19, Vite 8, Firebase Auth, Firestore, Vercel
Live Production Link	https://fitpulse-swe.vercel.app/

GitHub Repository Link	https://github.com/abhinavsai2006/fitpulse

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Academic Year 2025-2026

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1. Introduction

This report documents the complete implementation phase of FitPulse with emphasis on production-grade authentication, cloud-backed user data, and deployment stability. The coding work evolved from a UI-first prototype to a secured and deployable web application with Firebase integration.

1.1 Implementation goals

- Implement reliable user authentication with email/password and Google sign-in.
- Persist user profile and onboarding data in Firestore with secure access rules.
- Improve runtime resilience by handling popup, redirect, and profile-sync edge cases.
- Move environment-sensitive configuration outside source code and into environment variables.
- Deliver a production deployment that is verifiable through live and repository links.

2. Development Environment and Toolchain

Area	Tools and Configuration	Purpose
Frontend framework	React 19 with Vite 8 build pipeline	Fast development iteration and optimized production bundles
State management	Context API with reducer pattern	Predictable global state and action-based transitions

Area	Tools and Configuration	Purpose
Authentication	Firebase Authentication	Email/password, verification flow, reset email, Google sign-in
Data persistence	Cloud Firestore	User profile, onboarding, and app-related account data
Deployment	Vercel	Continuous hosting and production delivery
Source control	GitHub repository with authored commits	Version traceability and academic submission evidence

3. Architecture and Module Design

3.1 Frontend composition

The application is organized into pages, reusable components, data helpers, and a centralized AppContext. Screen transitions and data updates are driven by reducer actions to avoid inconsistent state mutations.

3.2 Firebase integration model

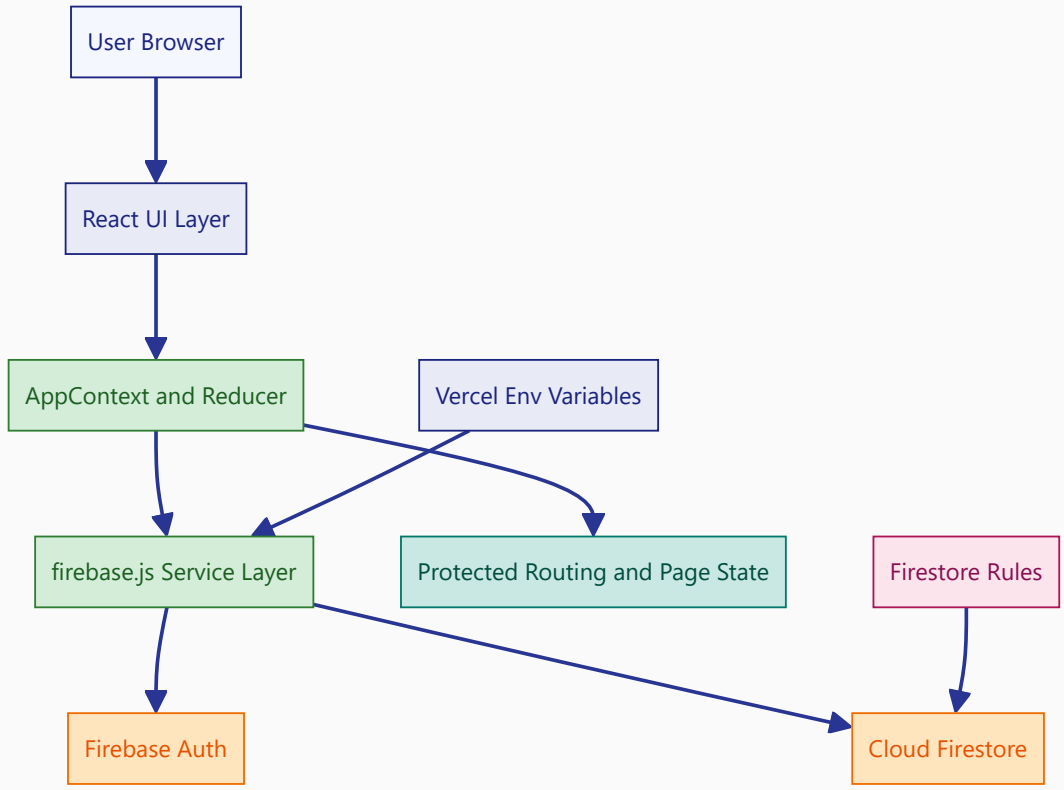

Figure 1: FitPulse frontend and Firebase integration architecture

4. Core Implementation Details

4.1 Authentication workflows

Flow	Implemented Behavior	Outcome
Email registration	Create account, send verification email, initialize profile, sign out until verified	Verified-account-first onboarding control
Email login	Authenticate credentials, block unverified accounts, resend verification	Reduced unauthorized incomplete-account sessions
Google sign-in	Popup-first strategy with redirect fallback for blocked/cancelled popup	Smoother login completion across browser restrictions
Password reset	Forgot-password trigger sends reset email from auth page	Self-service account recovery support
Logout	Sign out from Firebase and reset protected state	Clean session termination and route reset

4.2 State management and routing

The reducer in AppContext handles NAVIGATE, LOGIN, RESTORE_SESSION, LOGOUT, COMPLETE_ONBOARDING, START_WORKOUT, and END_WORKOUT actions. Route restoration is controlled by auth readiness and onboarding status, preventing landing-page loops after successful sign-in.

4.3 User profile and onboarding persistence

- ensureUserProfile creates or merges Firestore profile data on auth state changes.

- createUserViewModel merges profile values with safe defaults for UI rendering.
- completeOnboarding writes onboarding selections and stats to Firestore.
- Profile page now displays authenticated user data instead of static mock values.

4.4 Security and configuration hardening

Hardening Step	Implementation	Impact
Secret migration	Firebase config moved from source to VITE_FIREBASE_* environment variables	No sensitive deployment values in tracked source
Environment template	.env.example added with setup guidance	Improved reproducible setup for collaborators
Runtime config sanitization	Environment values normalized with trim before Firebase initialization	Prevents hidden whitespace misconfiguration
Rules enforcement	Firestore rules restrict read/write access to document owner	Protects user data from cross-account access

> **Production readiness note:**
> The implementation now includes secure configuration boundaries, owner-scoped database rules, and robust auth edge-case handling to support real-world usage.

5. Build, Quality, and Validation

Validation Item	Command or Method	Status
Production build generation	npm run build	Pass
Auth state restoration checks	Manual run across login/logout/refresh flows	Pass
Profile persistence checks	Firestore profile creation and update validation	Pass
Rule deployment validation	firebase deploy --only firestore:rules	Pass
Production redeploy	vercel deploy --prod --yes	Pass

6. Deployment and Release Operations

Operation	Description	Result
Vercel project alignment	Confirmed deployment scope under user account and linked project	Completed
Environment synchronization	VITE_FIREBASE_* values updated for Development and Production targets	Completed
Preview environment note	CLI requires branch-scoped preview variable setup in non-interactive mode	Documented
Firestore rules publishing	Rules deployed to fitpulse-7173f and confirmed active	Completed
Final production alias	Live URL verified as fitpulse-swe.vercel.app	Completed

7. Challenges and Resolutions

Challenge	Root Cause	Resolution
Auth flow bouncing to landing	Timing and state synchronization edge cases	Auth readiness and profile sync handling improved in AppContext
Google popup interruption	Popup blocked or cancelled in browser	Added popup-first flow with redirect fallback
Runtime config failures	Missing or malformed deployed environment values	Moved to env-based config and normalized values before initialization
Secret scanning alert	Firebase config present in tracked source	Migrated to environment variables and updated ignore/template files
Firestore access issues	Rules context mismatch during deployment and auth evolution	Standardized owner-only rules and redeployed to correct project

8. Key Deliverables

- Fully implemented multi-page React application with reducer-driven state architecture.
- Production-integrated Firebase Authentication (email/password + Google sign-in).
- Email verification enforcement and forgot-password recovery workflow.
- Cloud Firestore user-profile lifecycle with onboarding persistence.
- Secure Firestore rules and environment-variable based secret management.
- Deployment-ready build and live production verification on Vercel.
- Updated documentation artifacts for academic submission.

9. Conclusion

The implementation phase of FitPulse has progressed from UI-level development to a robust, cloud-integrated application suitable for demonstration and evaluation. Core coding objectives have been achieved with focus on maintainability, authentication reliability, data security, and deployment stability.

By combining modular frontend engineering with Firebase services and controlled deployment operations, the project now reflects professional software engineering practices expected in a complete academic submission.

9.1 Submission links

Resource	Link	Purpose
Live Production	https://fitpulse-swe.vercel.app/	Functional demonstration and final validation

Resource	Link	Purpose
GitHub Repository	https://github.com/abhinavsai2006/fitpulse	Source code and revision history evidence

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End of Document

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VIT-AP University | April 2026